

MATH3881/5881: Statistical Machine Learning Theory

Week 5: Theory Tutorial: Four Optimisers on a Constant Slope

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Week 4 introduced four parameter-update rules: **GD**, **Momentum**, **RMSProp**, and **Adam**. All four use a hyperparameter α usually called the “learning rate”. In this tutorial we ask the simplest possible question about all four:

If the gradient stops changing, what does one step look like, per coordinate?

The four answers split cleanly into two camps, and the split is exactly where the genuine algorithmic distinction lies.

Background

Notation. Write $\Delta\theta_i^{(t)} := \theta_i^{(t+1)} - \theta_i^{(t)}$ for the per-coordinate step at iteration t , and $g_i^{(t)} := \partial C / \partial \theta_i$ for the gradient component on coordinate i at that iteration. The four update rules expressing $\Delta\theta_i^{(t)}$ in terms of past gradients are the ones from the Week 4 lecture.

Constant-slope regime. Suppose the gradient on coordinate i stops changing: $g_i^{(t)} = g_i$ from some iteration on. For *large enough* t (a few iterations past the time constant of the running averages, not literally $t = \infty$), Momentum’s running average v_i and RMSProp’s running average s_i have effectively settled to constants:

$$v_i^{(t+1)} \approx g_i, \quad s_i^{(t+1)} \approx g_i^2.$$

Two problems

Problem 1: Two equivalences on a constant slope

Assume $|g_i| \gg \varepsilon$. Look up the update rules for GD, Momentum, RMSProp, and Adam in Week 4.

(a) Show that, for large t , the Momentum step on a constant slope is approximately equal to the GD step.

(b) Show that, for large t , the Adam step on a constant slope is approximately equal to the RMSProp step.

Problem 2: What kind of quantity is α ?

Adam and GD both call α a “learning rate.” They are not calling it the same thing. Consider a rescaling of coordinate i : define $\tilde{\theta}_i := \kappa \theta_i$ for some fixed $\kappa > 0$ (a change of units on that coordinate). Let $\tilde{C}(\tilde{\theta}) := C(\theta)$ and $\tilde{g}_i := \partial \tilde{C} / \partial \tilde{\theta}_i$.

- (a) Using the chain rule, show that $\tilde{g}_i = g_i / \kappa$.
- (b) Using the steady-state formulae from Problem 1, compute the per-coordinate step $\Delta \tilde{\theta}_i^{(t)}$ produced by GD and by RMSProp on the constant slope, in the new coordinates. Show that GD’s step is $1/\kappa$ times the original step, whereas RMSProp’s step is unchanged.
- (c) In one sentence, use the invariance you showed in (b) to explain why Adam’s default $\alpha = 10^{-3}$ ports across architectures whereas a good α for GD must be re-tuned for every new problem.

Why this matters

Different architectures place their parameters at very different natural scales: layer widths vary, initialisation schemes rescale weights, activation magnitudes differ across depths. The rescaling-invariance you established in (b) is exactly the property that lets a single $\alpha = 10^{-3}$ work out of the box on a dense classifier, a transformer, and a ResNet alike. GD’s non-invariance is exactly why it cannot.